

Byeongjee Kang

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Research Interests

I develop programming languages and compilers for writing safe and efficient programs on heterogeneous, resource-constrained devices. To this end, I work on optimal region splitting for intermittent computing, efficient program state migration, static resource analysis of programs, and WebAssembly compilation pipelines.

Professional Skills

Programming Languages: Rust, C++, WebAssembly, Python, SML, JavaScript, Scala, Maude, Futhark

Frameworks and Tools: LLVM, MLIR, Gurobi

Embedded Systems: MSP430, Arduino, bare-metal C, Otii Ace Pro (energy measurement), Saleae Logic (logic analyzer)

Education

PhD	Carnegie Mellon University , Software Engineering Advised by Prof. Limin Jia	Pittsburgh, USA Aug 2023 – present
BS	Pohang University of Science and Technology (POSTECH) , Computer Science and Engineering & Mathematics (double major) - GPA: 4.26/4.3 (Ranked 1st among all graduating students in class of 2023) - Entered with presidential scholarship offered to one outstanding high school student in the year, and graduated with the highest GPA - Early graduation by one-year (leave of absence for military duty during Jan 2020 – Dec 2021)	Pohang, Korea Feb 2018 – Feb 2023
Diploma	Korea Science Academy of KAIST , High School for Science-Gifted Students - GPA: 4.23/4.3 (Ranked 1st among all graduating students in class of 2018) - Funded by Korea Science Academy Fund Scholarship	Busan, Korea Feb 2015 – Feb 2018

Work Experiences

Microsoft Research , Research intern at Data Systems Group mentored by Kaushik Rajan - Compiler for database intrinsics: Auto-vectorizing compiler on MLIR with support for control flow dependencies - Developed compiler optimization passes for database intrinsics using MLIR and Polygeist, improving the performance of database intrinsic functions with control dependencies through vectorization	Redmond, WA, USA May 2025 – Aug 2025
Kodebox , Full-stack software engineer as military service alternative CodeChain Foundry: Open-source blockchain engine based on composable module system - Worked as core developer of consensus engine, implementing BLS signature aggregation scheme and designing interface for modules to communicate with consensus engine Zuzu: Web-based captable management software	Seoul, Korea Jan 2020 – Dec 2021

Publications

WAMI: Compilation to WebAssembly through MLIR without Losing Abstraction Byeongjee Kang, Harsh Desai, Limin Jia, Brandon Lucia 10.1145/3828170.3828172 (International Conference on Managed Programming Languages and Runtimes (MPLR) 2026) [code]	Jan 2026
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Fray: An Efficient General-Purpose Concurrency Testing Platform for the JVM

Jan 2025

Ao Li, Byeongjee Kang, Vasudev Vikram, Isabella Laybourn, Samvid Dharanikota, Shrey Tiwari, Rohan Padhye

[10.1145/3764119](#) (Object-oriented Programming, Systems, Languages, and Applications (OOP-SLA) 2025) [\[code\]](#)

GraFeyn: Efficient Parallel Sparse Simulation of Quantum Circuits

Jan 2024

Sam Westrick, Pengyu Liu, Byeongjee Kang, Colin McDonald, Mike Rainey, Mingkuan Xu, Jatin Arora, Yongshan Ding, Umut Acar

ieeexplore.ieee.org/document/10821338 (IEEE International Conference on Quantum Computing and Engineering (QCE) 2024 (Best Paper))

Narrowing and Heuristic Search for Symbolic Reachability Analysis of Concurrent Object-Oriented Systems

Jan 2024

Byeongjee Kang, Kyungmin Bae

[10.1016/j.scico.2024.103097](#) (Science of Computer Programming 2024) [\[code\]](#)

Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search

Jan 2022

Byeongjee Kang, Kyungmin Bae

dl.acm.org/doi/10.1145/3563822.3568017 (International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS) 2022) [\[code\]](#)

Under Review

Bao: Automatic Region Placement and Memory Allocation for Intermittent Computing

Byeongjee Kang, Jan Hoffmann, Limin Jia, Brandon Lucia, Feras Saad

[\[code\]](#)

Preprints

Mr3: An Execution Framework for Hive and Spark

Sungwoo Park, Seunggon Namgung, Byeongjee Kang

dx.doi.org/10.2139/ssrn.4587845

Talks

- An MLIR Dialect for WebAssembly, WebAssembly Workshop 2025, [\[recording\]](#)

Software

- **Bao**: LLVM toolchain that instruments programs for batteryless devices by solving a mixed-integer linear program
- **WAMI**: MLIR-based compilation pipeline for WebAssembly
- **OCaMI**: Compiler from a functional subset of OCaml to WebAssembly through WAMI
- **rustlane**: ISPC-style SPMD programming for Rust, lowering scalar control flow to masked SIMD
- **Fray**: Controlled concurrency testing framework for the JVM
- **heuristic-narrowing-search**: Maude implementation of narrowing-based symbolic reachability analysis
- **CodeChain Foundry**: Open-source blockchain engine based on composable module system

Ongoing Research Projects

- **Automatic Program Rescaling and Migration**: Programming language and runtime support for automatically rescaling programs for resource-constrained systems
- **Probabilistic Modeling and Analysis Tools for Intermittent System**: Probabilistic modeling of resource usage in intermittent systems

Honors and Scholarships

ILJU Fellowship (ILJU Academy and Culture Foundation)	Aug 2023 – July 2028
Received total of USD 120,000 over eight semesters during graduate studies	
Program for Highly Dedicated Students (POSTECH)	Feb 2018 – Feb 2023
Presidential scholarship and supporting program in POSTECH offered to one outstanding freshman each year	
Presidential Science Scholarship (Ministry of Education of Korea)	Feb 2018 – Feb 2023
National science scholarship offered to top freshmen students in the field of science and technology	